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Independent bit error rate test device

Abstract

An electronic device is provided for conducting an independent bit error rate (BER) test to measure signal noise from a communication system that receives a radio frequency (RF) signal through a power combiner. The device includes a demodulator, a test bit pattern generator, a bit pattern comparator, a modulator and a computation processor. The demodulator receives the RF signal. The pattern generator provides a set bit sequence pattern. The bit pattern comparator compares the sequence pattern and the RF signal as a comparative signal. The modulator provides the sequence pattern from the generator as a carrier signal to the power combiner. The computation processor calculates BER from the comparative signal. In alternative embodiments, the generator, modulator, demodulator and comparator are combined as an efficient modem.

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Background/Summary

BACKGROUND

(1) The invention relates generally to satellite communication. In particular, the invention relates to interference assessment regarding communication via orbiting satellite relays.

(2) FIG. 1 shows a schematic view **100** of a conventional satellite communication link. An orbiting relay satellite **110** provides a communications interface between a transmitting gateway **120** and a receiving subscriber **130**, both on or near earth's surface. The gateway **120** emits an uplink radio frequency (RF) signal **140** to the satellite **110**, which in turn provides a downlink RF signal **150** to the subscriber **130**.

(3) Satellite Communication (SATCOM) Systems depend on a Radio Frequency (RF) transceiver on an orbiting satellite **110** to receive an uplink from one earth station (gateway) **120** and re-transmit it through a downlink to a subscriber **130** to exchange data in view **100**. This process is also performed in reverse to send data from the subscriber **130** back to the gateway **120**. There are many factors which can degrade the integrity of the data as the signals travel from the gateway **120** to the subscriber **130**.

(4) FIG. 2 shows a schematic view **200** of a conventional satellite communication link. As with view **100**, communication commences between the gateway **120** and the subscriber **130** via the satellite **110**. The gateway **120** transmits an upload data packet **210** the satellite **110** that in turn forwards the data packet **220** to the subscriber **130** to as received data **240** and associated bit error **250**.

(5) Conducting a bit error rate test (BERT) between the gateway **120** and the subscriber **130** in view **200** constitutes a common technique to evaluate the quality of the link. The BERT operates by generating a known data sequence or pattern at the gateway **120**, transmitting it over the link to the subscriber **130**, and comparing it to the expected pattern. Information bits that are received incorrectly are known as errors, and the bit error rate (BER) describes the ratio of error bits to total

bits sent.

SUMMARY

(6) Conventional bit error tests yield disadvantages addressed by various exemplary embodiments of the present invention. In particular, various exemplary embodiments provide an electronic device for conducting an independent bit error rate (BER) test to measure signal noise from a communication system that receives a radio frequency (RF) signal through a power combiner. The device includes a demodulator, a test bit pattern generator, a bit pattern comparator, a modulator and a computation processor. The demodulator receives the RF signal. The pattern generator provides a set bit sequence pattern. The bit pattern comparator compares the sequence pattern and the RF signal as a comparative signal. The modulator provides the sequence pattern from the generator as a carrier signal to the power combiner. The computation processor calculates BER from the comparative signal. In alternative embodiments, the generator, modulator, demodulator and comparator are combined as an efficient modem.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and various other features and aspects of various exemplary embodiments will be readily understood with reference to the following detailed description taken in conjunction with the accompanying drawings, in which like or similar numbers are used throughout, and in which:
- (2) FIG. 1 is a schematic view of a conventional satellite link;
- (3) FIG. 2 is a schematic view of a conventional over-the-air bit error rate test (BERT);
- (4) FIG. 3 is a schematic view of an external interferer in a subscriber system;
- (5) FIG. 4 is a schematic view of an off-satellite subscriber BERT;
- (6) FIG. 5 is a block diagram view of the off-satellite subscriber BERT;
- (7) FIG. 6 is a graphical diagram view of satellite downlink with potential test channels;
- (8) FIG. 7 is a graphical view of satellite downlink measurement techniques;
- (9) FIG. 8 is a block diagram view of an unmanned BERT radio frequency hardware diagram; and
- (10) FIG. 9 is a flowchart diagram view of a BERT configuration.

DETAILED DESCRIPTION

(11) In the following detailed description of exemplary embodiments of the invention, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration specific exemplary embodiments in which the invention may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments may be utilized, and logical, mechanical, and other changes may be made without departing from the spirit or scope of the present invention. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope of the present invention is defined only by the appended claims.

(12) In accordance with a presently preferred embodiment of the present invention, the components, process steps, and/or data structures may be implemented using various types of operating systems, computing platforms, computer programs, and/or general purpose machines. In addition, artisans of ordinary skill will readily recognize that devices of a less general purpose nature, such as hardwired devices, may also be used without departing from the scope and spirit of the inventive concepts disclosed herewith. General purpose machines include devices that execute instruction code. A hardwired device may constitute an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), digital signal processor (DSP) or other related component.

(13) The disclosure generally employs quantity units with the following abbreviations: length in meters (m), time in seconds(s), frequency in hertz (Hz), power in decibels (dBm), angles in

degrees) ($^{\circ}$), force in newtons (N), temperature in kelvins (K) and energy in joules (J). Supplemental measures can be derived from these, such as density in grams-per-cubic-centimeters (g/cm.sup.3), moment of inertia in kilogram-square-meters (kg-m.sup.2) and the like.

(14) Exemplified embodiments provide an independent or unmanned Bit Error Rate Test (UBERT) to conduct an automated Satellite Communication (SATCOM) electromagnetic interference (EMI) Test. UBERT can be conducted without requiring distant-end support and coordination with a remote gateway **120**, a valid satellite assignment, or manual determination of frequency or power settings.

(15) FIG. **3** shows a schematic view **300** of a satellite communication link that experiences interference. A naval combat ship **310** (e.g., aircraft carrier) includes a separate transmitter **320** that emits RF interference **330**. Meantime, the gateway **120** transmits to the subscriber **130** via the satellite **110**, while the subscriber **130** is subject to the RF interference **330**.

(16) In some SATCOM system installations, the suspected source of degradation can be isolated to EMI caused by an external RF emitter **320** operating in close proximity to the subscriber antenna system **130** in view **300**. In these scenarios, one factor that determines the level of degradation is the pointing angle of the SATCOM antenna. Therefore, one would preferably steer the antenna of the subscriber **130** manually away from the satellite **110** during a controlled test event to evaluate the degradation as the pointing angle changes.

(17) FIG. **4** shows a schematic view **400** of an off-satellite communication link that experiences interference. This includes a SATCOM transmitter **410**, an RF combiner **320** with transmitted data **230** and a SATCOM receiver **430** with received data **240** with the error bit **250** for the bit error rate test (BERT). The RF interference **330** from the transmitter **320** reaches the subscriber **130**, which passes signal interference **440** to the combiner **420**, which also receives a transmit signal **450** from the SATCOM transmitter **410**. The combiner **420** joins the signals **440** and **450** as a combination signal **460** to the SATCOM receiver **430**. The subscriber **130** includes an antenna **470** for receiving signals **150** and **330**.

(18) In order to perform a bit error rate test (BERT) while the subscriber **130** is not connected to a satellite **110**, a local BERT can be used shown in view **400**. In this arrangement, the SATCOM transmitter **410** replaces the downlink **150** from the satellite **110** and is combined with the RF interference **330** to be received by the subscriber antenna **470**.

(19) FIG. **5** shows a block diagram view **500** of an off-satellite communication link that experiences interference. A via-sat modern internal UBERT module **510** processes signals, both RF and intermediate frequency (IF). The BERT **510** includes a demodulator **515**, a test bit pattern generator **520**, a bit pattern comparator, a modulator **530** and a bit error rate (BER) computer **535**. The generator **520** provides a set bit sequence as a standard for comparison. The comparator **525** receives signals from the demodulator **515** and generator **520**, which also sends comparative signals to the modulator **530**. The BER computer **535** receives these comparative signals from the comparator **525**.

(20) The UBERT **510** communicates with an RF-to-IF downconverter **540** and a power combiner **550**. The downconverter **540** receives an interference signal **560** from the subscriber **130** and passes that receiver signal **570** to the power combiner **550**, which transmits a combined signal-and-interference signal **580** to the demodulator **515**. The combiner **550** receives a SATCOM test signal carrier **590** from the modulator **530** in response to the generator **520**.

(21) This test illustrated in view **500** shows implementation with a Viasat MD-1366 Enhanced Bandwidth Efficient Modem (EBEM) that combines the modulator **530** and demodulator **515**. The EBEM (**880** in FIG. **8**) conveniently provides the BERT pattern generator **520**, modulator **510** and demodulator **515**, and comparator **525** in a consolidated UBERT device **510**.

(22) Description and operation. The exemplary Unmanned BERT (UBERT) system includes the UBERT device **510** associated with the downconverter **540** and power combiner **550**. This exemplary UBERT system is intended to operate while the subscriber system **130** connects to the

satellite **110**. This eliminates the need for the conventional controlled test event as previously described, which is labor intensive and prohibits normal SATCOM system operation for the duration of the error test. Additionally, unlike the over-the-air BERT shown in view **200**, the UBERT system does not depend on the gateway **120** or satellite services to conduct an error test. (23) This is achieved by using a BERT configuration similar to that shown in view **400**, but with the SATCOM transmitter **410** configured to transmit on a different channel than that assigned to the subscriber **130** by the satellite provider. This permits the subscriber **130** to remain connected to the satellite **110** and maintain a communication link **150** using its assigned channel as the test signal carrier **590**.

(24) FIG. **6** shows a graphical view **600** of a satellite downlink plot. Frequency **610** (MHz) denotes the abscissa, while power **620** (dBm) denotes the ordinate. A signal trace **630** includes subscriber downloads **640** at approximately between -30 dBm to -35 dBm and UBERT test channels **650** at -45 dBm that represent noise.

(25) FIG. **7** shows a graphical view **700** of a signal measurement plot. Frequency **710** (MHz) denotes the abscissa, while power **720** (dBm) denotes the ordinate. A legend **730** identifies separate traces as mode **740** and mean **750**, with the latter being about 3 dBm higher at the UBERT levels than the former. View **700** illustrates an example of the scenario where the lowest portions of the mean trace **750** have been elevated by interference, while the mode trace **740** reports the correct values. This is especially important while determining the transmit power necessary for the test, as the trace obtained with the mean trace **750** would result in testing with too high of a power level. This key change greatly improved the accuracy of the test for those types of interference and was adopted as the primary measurement technique for all circumstances.

(26) FIG. **8** shows a block diagram view **800** of an exemplary RF hardware schematic **810** for the UBERT **510**. This includes an input **580**, a playback device **820**, a signal splitter **830**, a -10 dB signal combiner **840**, a minicircuit RF switch box **850** with switches A **852** and B **854**, spectrum analyzer **860**, a -10 dB signal divider **870**, an enhanced bandwidth efficient modem (EBEM) **880** with receive and transmit electrodes, and a sample port **880** for a channel, such as the test signal carrier **590**.

(27) The exemplary Unmanned BERT system addresses two additional problems which arise with this new test methodology. First, UBERT **510** uses a RF spectrum analyzer to characterize the downlink and determine the location of vacant channel appropriate for testing in view **600**. This is accomplished using an algorithm that compares the relative flatness and amplitude of potential channels; the optimal test channel has a lowest amplitude and flattest profile.

(28) Second, the Unmanned BERT system must determine the transmit power level used to simulate the subscriber downlink. The UBERT **510** must determine a realistic level, because too high of a power level will fail to detect degradation and too low of a level will report degradation too often. This operated is accomplished using a spectrum analyzer **860**, which calculates the power received in the test channel over a short period to determine the noise floor of that channel **865**.

(29) All components of the exemplary Unmanned BERT system utilize commercial off-the-shelf (COTS) components connected in hardware **810** that are controlled by software written in MATLAB. The primary components are the Viasat MD-1366 EBEM **880**, which provides both the SATCOM transmitter and receiver functions using an embedded BERT test mode, and a Tektronix RSA306, which is a compact universal serial bus (USB) real-time spectrum analyzer **860** capable of performing the required RF measurements through an application programming interface (API). The RF Hardware arrangement is shown in view **800**.

(30) FIG. **9** shows a flowchart diagram view **900** of an unmanned BERT software flow diagram that begins at start **905** within a MATLAB terminal program **910** referred to as software. Start **905** initiates receipt by an import extensible markup language (XML) configuration **915** of input data **580** by a BERT configuration builder **920** that formats the data by XML BERT configuration **925**.

The data in the builder **920** include select frequencies, waveforms and bit error rate test options. An increment BERT index **930** receives the data from the configuration **920** to determine RSA alignment **932** and to reorganize IF/IQ files **934**. Note that IQ data includes projection and phase.

(31) A first playback decision query **936** determines whether to replay the data. A negative response continues to input level check **938**, whereas a positive response initiates IQ playback **940** before resuming to the check **938** that proceeds to a fixed frequency query **942**. A positive response sets EBEM IF frequency **944**, whereas a negative response diverts to locating the frequency **946** before resuming to the set frequency operation **944** before continuing to measuring RF losses **948**.

(32) This measurement from operation **948** leads to a power query **950** for fixed transmission. A positive response sets EBEM IF transmission level **952**, while a negative response diverts to determine IF transmission level **954**. From the set level operation **952**, the process continues to waiting for carrier lock **956** and proceeding to a second playback query **958**. A negative response initiates a lead-in pause **960**. A positive response diverts the process to restart IQ playback **962** before resuming to the lead-in pause **960** that proceeds to starting IF/IQ stream and BERT operation **964**. The process waits for BERT completion **966** before reaching a histogram query **968**. The stream and BERT operation **964** also retrieves an R3F or TIQ file **970** to create an IQ histogram **972**, both of which then write to an IF/IQ file directory **974**. The histogram **972** is written in comma separated values (CSV) format.

(33) The create histogram operation **972** also saves BERT results **976** as CSV. A positive response to the histogram query **968** proceeds to the create histogram **972** operation, whereas a negative response resumes to the save results **976** in a program directory **978**. The operation proceeds to purge IQ data **980**, following the save operation **976**. The operation proceeds to wait for specified BERT spacing delay **985** before ending **990**. Flowchart view **900** further includes a note **995** of batch script to call MATLAB in a perpetual loop.

(34) Software Walkthrough: The software **910** controlling the exemplary Unmanned BERT system is illustrated as the flow chart in view **900** as previously described. In order to script what type of test is conducted the operator must define a configuration file using the “BERT Configuration Builder” **920**.

(35) The process begins **905** by importing the builder file **920** to learn what type of test is to be performed. Next, the spectrum analyzer **860** undergoes an alignment routine that requires the RF switches **852** and **854** to move to a disconnected state. Following alignment, the software **910** determines whether there is enough file space to store a digital recording of the RF energy. If not, the software **910** moves files as needed to external data storage.

(36) If the test configuration as query **936** is set for lab playback, the software initiates file playback **940** via an application programming interface (API) with a COTS RF playback device **820** connected to the UBERT system input **580**. Next, the software **910** uses the spectrum analyzer **860** to measure the power level at the input **580** of the UBERT system to ensure satisfaction above the sensitivity threshold of the spectrum analyzer **860**; thus providing valid measurement values.

(37) If the configuration is set to locate a vacant SATCOM channel (as opposed to a fixed frequency) as query **942**, the spectrum analyzer **860** acquires the trace and the software **910** computes moving mean and moving standard deviation values to determine the best channel for testing. To ensure that the modem **880** is still properly connected to the UBERT system, the software next executes two RF loss measurements **948** to determine the loss between the modem transmitter and receiver ports, and between the modem transmitter and spectrum analyzer ports.

(38) If the configuration **925** calls for dynamic setting of the transmit power level as query **950**, the software **910** proceeds with measuring the mode of the selected SATCOM channel and using a lookup table to achieve the desired signal-to-noise ratio at the receiver on the modem **880**. This can be accomplished even in the presence of pulsed interference due to measuring the statistical mode of the channel. Once the transmit level is determined as query **950**, the software **910** re-starts the playback **962** if needed as query **958**, and then commands the modem **880** to begin a built-in BERT

test sequence.

(39) Simultaneously with the UBERT **510**, the spectrum analyzer **860** is commanded to capture the interference waveform—without the SATCOM transmitted signal **410**—for later analysis or playback use **940**. At the conclusion of the test, the captured RF data **976** are either immediately analyzed, saved to long-term storage **978**, or deleted **980** if the observed BER is not noteworthy. Deleting captured data for passing BER tests is beneficial to reduce the total amount of storage required for a complete test event.

(40) The exemplary Unmanned BERT (UBERT) system has commercial potential for any satellite systems that routinely experiences degradation due to EMI at their subscriber **130** location. Due to the use of the Viasat MD-1366 modem **880**, the exemplary UBERT system is targeted for military applications and was developed to specifically address EMI concerns aboard the USS Gerald R. Ford (CVN 78). Due to the limited time available for dedicated EMI testing aboard CVN 78, a system that can autonomously collect accurate EMI measurements on a not-to-interfere basis was extremely attractive.

(41) Exemplary embodiments provide various advantages over conventional techniques for signal. The Unmanned BERT system has an advantage over other test systems and methodologies because it can be deployed on an existing SATCOM system and performs accurate EMI measurements without interfering with SATCOM operations. Additionally, automation enables these tests to be performed without manual interaction that makes obtaining these measurements more cost effective.

(42) Using a spectrum analyzer **860** to collect traces weighted by their statistical mode is an innovative approach to locating a vacant downlink channel, despite the presence of significant pulsed interference, for the purpose of conducting a BERT **510** to evaluate downlink interference **150**. Additionally, using the same measurement technique to accurately set the transmit power level of the BERT **510** is also a novel and effective approach. The Spectrum Analyzer integrated with the Unmanned BERT system also provides the ability to digitally record the interference during the test, which adds a capability of indefinite playback in a lab environment. The Unmanned BERT software **910** supports this capability by enabling the operator to specify files for playback and automatically repeats the BERT **510** using the same algorithms as though it were running a field test.

(43) One aspect of the exemplary process includes obtaining the RF power traces to both locate a vacant channel and determine the appropriate transmit power level. Prior to this, the spectrum analyzer **860** was configured to measure an “average trace” from which to compute the statistical mean of approximately one-hundred individual traces to determine the noise level without being influenced by pulsed interference. This proved ineffective for some types of interference, so the solution was to calculate the statistical mode, rather than mean, which was less influenced by high duty-cycle interference. This measurement technique is not a common feature of spectrum analyzers and requires an additional calculation in the unmanned BERT software to implement.

(44) Alternatives to the Unmanned BERT system include other methods that decode the satellite downlink and predict BERs. These can be very useful and accurate, but are limited to only assessing the conventional satellite assignment. The exemplary system can locate test channels anywhere within the receive frequency range and perform tests at those locations. Another alternative is to manually perform BERTs, but this is labor-intensive and often unfeasible for extended test periods.

(45) On-Air Testing: Conventional testing requires terminating the data link (disrupting service) and moving into a test mode. By contrast, the exemplary UBERT **510** enables the user's data link to remain connected and tests an appropriate neighboring channel. This channel is exposed the same interference conditions to be evaluated as the data link for communication. UBERT **510** couples the test signal from the subscriber **130** into the same path that receives interference **330**.

(46) Automatic Channel Detection and Level Setting: The algorithms within UBERT **510**

automatically measure the spectrum and determine the optimal RF channel to perform the test. Additionally, the power level at which to run the test is determined automatically—this is a critical aspect of obtaining meaningful results. UBERT **510** accomplishes this with a digital signal processing algorithm to measure the RF energy and calculate the required power levels.

(47) Data Collection Automation: All data, to include both test logs and high-fidelity recordings of the spectrum when anomalous conditions are detected, is automatically created and retained for in-depth analysis at the end of the event. UBERT **510** accomplishes this with an algorithm that compares measured performance of the bit-error-rate test to a user-configurable threshold. Under circumstances in which the performance does not satisfy the threshold, the data are automatically archived by the system for later analysis.

(48) While certain features of the embodiments of the invention have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the embodiments.

Claims

1. An electronic device for conducting an independent bit error rate (BER) test to measure signal noise from a communication system that receives a radio frequency (RF) signal through a power combiner, said device comprising: a demodulator for receiving the RF signal from the power combiner and producing a demodulated signal; a test bit pattern generator for providing a set bit sequence pattern; a bit pattern comparator for comparing said sequence pattern and demodulated signal as a comparative signal; a modulator for providing said sequence pattern from said generator as a carrier signal to the power combiner; and a computation processor for calculating BER from said comparative signal.
 2. The device according to claim 1, wherein said generator, modulator, demodulator and comparator are combined as an efficient modem.
 3. The device according to claim 2, further comprising: a playback device for receiving the RF signal; a signal splitter for splitting the RF signal into first and second input signals; a switch circuit having first and second switches, said first switch for receiving said first input signal and a third input signal to provide said comparative signal, and said second switch for receiving said second input signal to provide said third input signal and a fourth input signal; a signal combiner for combining said second input signal with said carrier signal into a feedback signal to the power combiner; a spectrum analyzer for receiving said comparative signal for determining received signal power; and a signal divider for providing said fourth input signal to said efficient modem.
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